

\text{Scaled Gradient} = 0.00005 \times 2,000.0 = \mathbf{0.1}7th Meeting 12/9

1. Explain gradient 1 / loss function value.

KENDAL function using Uncertainty to Weigh Losses. Start with softmax

$$\begin{aligned}
&= -\log p(\mathbf{y}_1, \mathbf{y}_2 = c | \mathbf{f}^{\mathbf{W}}(\mathbf{x})) \\
&= -\log \mathcal{N}(\mathbf{y}_1; \mathbf{f}^{\mathbf{W}}(\mathbf{x}), \sigma_1^2) \cdot \text{Softmax}(\mathbf{y}_2 = c; \mathbf{f}^{\mathbf{W}}(\mathbf{x}), \sigma_2) \\
&= \frac{1}{2\sigma_1^2} \|\mathbf{y}_1 - \mathbf{f}^{\mathbf{W}}(\mathbf{x})\|^2 + \log \sigma_1 - \log p(\mathbf{y}_2 = c | \mathbf{f}^{\mathbf{W}}(\mathbf{x}), \sigma_2) \\
&= \frac{1}{2\sigma_1^2} \mathcal{L}_1(\mathbf{W}) + \frac{1}{\sigma_2^2} \mathcal{L}_2(\mathbf{W}) + \log \sigma_1 \\
&\quad + \log \frac{\sum_{c'} \exp\left(\frac{1}{\sigma_2^2} f_{c'}^{\mathbf{W}}(\mathbf{x})\right)}{\left(\sum_{c'} \exp\left(f_{c'}^{\mathbf{W}}(\mathbf{x})\right)\right)^{\frac{1}{\sigma_2^2}}} \\
&\approx \frac{1}{2\sigma_1^2} \mathcal{L}_1(\mathbf{W}) + \frac{1}{\sigma_2^2} \mathcal{L}_2(\mathbf{W}) + \log \sigma_1 + \log \sigma_2, \\
&\quad \text{where again we write } \mathcal{L}_1(\mathbf{W}) = \|\mathbf{y}_1 - \mathbf{f}^{\mathbf{W}}(\mathbf{x})\|^2 \quad (10) \\
&\quad \text{for the Euclidean loss of } \mathbf{y}_1, \text{ write } \mathcal{L}_2(\mathbf{W}) =
\end{aligned}$$

But for my model, I use gaussian:

Function:

$$P(y|x, w) = \mathcal{N}(y; f(x), \sigma^2)$$

- **P**: "Probability"
- **I** : "Given" (conditional bar)
- **N** : "Normal" (referring to the distribution)

- **f(x)**: "f of x" (the mean/prediction)
- **σ²**: "Sigma squared" (the variance)

The full Gaussian Probability Density Function (PDF):

$$P(y|x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{\|y - f(x)\|^2}{2\sigma^2}\right)$$

To train a model, we want to Maximize Probability (Likelihood), which is mathematically the same as **Minimizing Negative Log-Likelihood (NLL)**.

$$NLL = -\log\left(\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{\text{Loss}}{2\sigma^2}\right)\right)$$

The Expansion

Using log rules ($\log(a \cdot b) = \log a + \log b$):

$$NLL = -\log\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right) - \log\left(\exp\left(-\frac{\text{Loss}}{2\sigma^2}\right)\right)$$

Simplify the exponential ($\log(\exp(x)) = x$)

$$NLL = -\log\left((2\pi\sigma^2)^{-1/2}\right) - \left(-\frac{\text{Loss}}{2\sigma^2}\right)$$

Simplify the first term

$$NLL = \frac{1}{2} \log(2\pi\sigma^2) + \frac{1}{2\sigma^2} \text{Loss}$$

expand the log

$$\log(2\pi\sigma^2) = \log(2\pi) + \log(\sigma^2) = \text{Constant} + 2 \log \sigma$$

$$NLL = \underbrace{\frac{1}{2\sigma^2} \text{Loss}}_{\text{Weighted Error}} + \underbrace{\log \sigma}_{\text{Regularizer}} + \underbrace{\text{Constant}}_{\text{Ignored}}$$

$$\mathcal{L}_{total} = \underbrace{\frac{1}{2\sigma_{det}^2} \mathcal{L}_{det}}_{Detection} + \underbrace{\frac{1}{2\sigma_{pose}^2} \mathcal{L}_{pose}}_{Pose} + \underbrace{\frac{1}{2\sigma_{act}^2} \mathcal{L}_{act}}_{Activity} + \underbrace{\log(\sigma_{det}) + \log(\sigma_{pose}) + \log(\sigma_{act})}_{Regularization}$$

Gradient function:

$$J(\theta) = -\ln(P(\theta))$$

$$\frac{dJ}{d\theta} = \frac{dJ}{dP} \times \frac{dP}{d\theta}$$

$$\frac{dJ}{dP} = \frac{d}{dP}[-\ln(P)] = -\frac{1}{P}$$

$$\nabla J = \left(-\frac{1}{P}\right) \times \nabla P$$

$$\text{Gradient} \propto \frac{1}{P}$$

Loss is proportional to Probability

$$Loss(L) = -\ln(P)$$

example:

1. Perfect prediction

$$P = 1.0 : L = -\ln(1) = 0$$

2. Bad prediction

$$P = 0.1 : L = -\ln(0.1) \approx 2.3$$

3. Terrible prediction

$$P = 0.001 : L = -\ln(0.001) \approx 6.9$$

The explanation without weighted loss function

Pose Loss is **0.0005**

$$\text{Gradient} \propto \frac{1}{\text{Loss}}$$

LR: 0.001

$$\text{Gradient} = \frac{1}{0.0005} = \frac{1}{5 \times 10^{-4}} = \mathbf{2,000.0}$$

Weight Update

$$\text{Update} = \text{Gradient} \times \text{LR} = 2,000.0 \times 0.001 = \mathbf{2.0}$$

update model weight

$$W_{\text{new}} = W_{\text{old}} - \text{Update} = 0.01 - 2.0 = \mathbf{-1.99}$$

Forward Pass

$$Z = W_{\text{new}} \times \text{Input} = -1.99 \times 500 \approx \mathbf{-1,000}$$

There will be function failure

$$\text{Value} = e^{-Z} = e^{1000}$$

$$e^{1000} \approx \text{Infinity}$$

$$\text{Result} = \frac{\infty}{\infty} = \text{NaN}$$

Conclusion: The tiny number **0.0005** started a chain reaction that resulted in **Infinity**, crashing the training.

Softmax Formula:

$$P_i = \frac{e^{Z_i}}{\sum_j e^{Z_j}}$$

$$\text{Result} = \frac{\text{Numerator}}{\text{Denominator}} = \frac{\infty}{\infty}$$

What if I use Kendall loss?

$$\text{Weight} = \frac{1}{2\sigma^2} = \frac{1}{2(100)^2} = \frac{1}{20,000} = \mathbf{0.00005}$$

Why the sigma is 100?, assume the total gradient (0.1 [safe])

$$\text{Total Gradient} = \frac{1}{2\sigma^2} \times \text{Raw Gradient}$$

$$0.1 = \frac{1}{2\sigma^2} \times 2000$$

$$0.2\sigma^2 = 2000$$

$$\sigma^2 = \frac{2000}{0.2} = 10,000$$

$$\sigma = \sqrt{10,000} = \mathbf{100}$$

Back to topic:

$$\text{Scaled Gradient} = \text{Weight} \times \text{Raw Gradient}$$

$$\text{Scaled Gradient} = 0.00005 \times 2,000.0 = \mathbf{0.1}$$

$$\text{Update} = \text{Scaled Gradient} \times \text{LR} = 0.1 \times 0.001 = \mathbf{0.0001}$$

$$W_{\text{new}} = 0.01 - 0.0001 = \mathbf{0.0099}$$

How did they adjust the value of this gradient (in the paper)

$$L(\sigma) = \frac{1}{2\sigma^2} \text{Loss}_{\text{raw}} + \log(\sigma)$$

term 1

$$\frac{d}{d\sigma} \left(\frac{1}{2} \sigma^{-2} \cdot \text{Loss} \right) = \frac{1}{2} (-2\sigma^{-3}) \cdot \text{Loss} = -\frac{\text{Loss}}{\sigma^3}$$

term 2

$$\frac{d}{d\sigma} (\log \sigma) = \frac{1}{\sigma}$$

so

$$\frac{\partial L}{\partial \sigma} = -\frac{\text{Loss}}{\sigma^3} + \frac{1}{\sigma}$$

If it has any improvement → model

I will breakdown the model architectural workernet (what is the improvement of this mdoel → compare to the base model)

Write the mathematical equation function loss (explain the normal loss function)

I will research about the conference that matched my project → look wiki cfp call for paper (for asia only)